Bayesian Inference for AI: A Comprehensive Guide to Bayesian Networks, Naïve Bayes and Hidden Markov Models

Bayesian Network

A *Bayesian Network* is a directed acyclic graph (DAG) that represents the probabilistic relationships among a set of random variables. Each node in the graph corresponds to a random variable, and the edges represent the conditional dependencies between these variables.

Formally, a Bayesian Network for a set of random variables $X_1, X_2, \ldots, X_n$ consists of:

- A set of *nodes* representing the variables $X_1, X_2, \ldots, X_n$,
- A set of *directed edges* that encode conditional dependencies between the variables,
- A set of *conditional probability distributions* $P(X_i \mid \text{Parents}(X_i))$, where $\text{Parents}(X_i)$ represents the set of parents of X_i in the network.

1 Joint Probability Distribution

The *joint probability distribution* of all the variables in a Bayesian Network can be factored as the product of the conditional probabilities of each variable given its parents in the network. Mathematically, the joint probability distribution can be written as:

$$P(X_1, X_2, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid \text{Parents}(X_i))$$

This factorization is based on the assumption that each variable is conditionally independent of its non-descendants given its parents.

Example

Consider a Bayesian Network with three variables: X_1, X_2, and X_3, where:

- X_1 has no parents,
- X_2 has X_1 as its parent,
- X_3 has X_1 and X_2 as its parents.

The joint probability distribution for this network is:

$$P(X_1, X_2, X_3) = P(X_1) \cdot P(X_2 \mid X_1) \cdot P(X_3 \mid X_1, X_2)$$

Let's work through an example similar to the alarm for burglary and earthquake scenario, where we will calculate everything from scratch, including the dataset, conditional probabilities, the Bayesian Belief Network and how to perform inference using Conditional Probability Tables (CPTs). A small historical database that contains simulated data for burglary, earthquake, alarm activation, and whether John and Mary called. We will use this data to calculate the prior probabilities and conditional probabilities needed for the Bayesian Belief Network.

2 Events

- *Burglary (B)*: There is a burglary at your house.
- *Earthquake (E)*: An earthquake occurs.
- *Alarm (A)*: The alarm goes off.
- *JohnCalls (J)*: John calls if he hears the alarm.
- *MaryCalls (M)*: Mary calls if she hears the alarm.

3 Bayesian Network Graph

The following is the graphical representation of the Bayesian Network for this scenario:

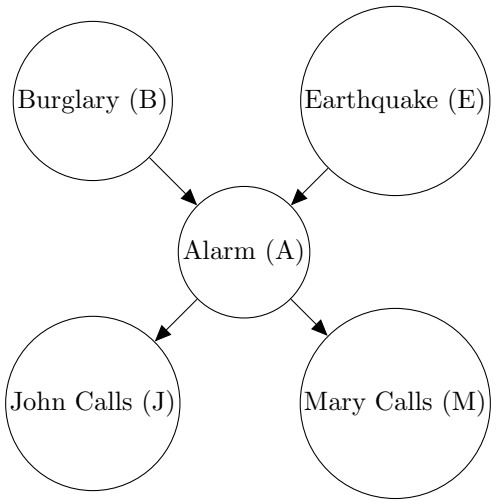


Figure 1: Bayesian Network

4 Problem Statement

We are tasked with finding the probability of the event where the alarm has sounded (A = Yes), a burglary has occurred (B = Yes), an earthquake has not occurred (E = No), and both John and Mary have called (J = Yes, M = Yes). We will calculate the joint probability of this event using a Bayesian network model based on the dataset provided.

5 Historical Database

Below is the historical data showing occurrences of burglary, earthquake, alarm activations, and calls from John and Mary:

6 Prior and Conditional Probabilities

From the dataset, we can calculate the following prior and conditional probabilities:

6.1 Prior Probabilities

- $P(B = \text{Yes}) = \frac{4}{10} = 0.4$
- $P(B = \text{No}) = \frac{6}{10} = 0.6$
- $P(E = \text{Yes}) = \frac{5}{10} = 0.5$
- $P(E = \text{No}) = \frac{5}{10} = 0.5$

Table 1: Historical dataset showing Burglary, Earthquake, Alarm, John Calls, and Mary Calls

Event ID	Burglary (B)	Earthquake (E)	Alarm (A)	John Calls (J)	Mary Calls (M)
1	Yes	No	Yes	Yes	Yes
2	No	Yes	Yes	Yes	No
3	Yes	Yes	Yes	Yes	Yes
4	No	No	No	No	No
5	Yes	No	Yes	No	Yes
6	No	Yes	No	No	No
7	No	No	Yes	Yes	No
8	Yes	Yes	Yes	Yes	Yes
9	No	No	No	No	No
10	No	Yes	Yes	Yes	Yes

6.2 Conditional Probability Tables (CPT)

Next, we calculate the conditional probabilities for the alarm going off given different combinations of burglary and earthquake, and the likelihood of John and Mary calling given that the alarm went off or didn't go off.

6.2.1 Conditional Probability of Alarm $P(A|B, E)$:

- **Case 1:** $P(A = \textbf{Yes}|B = \textbf{Yes}, E = \textbf{Yes})$**:**
  - Number of cases where $B =$ Yes and $E =$ Yes: 2.
  - Out of these, the alarm went off in all 2 cases.

  Thus, we can calculate:
  $$P(A = \text{Yes}|B = \text{Yes}, E = \text{Yes}) = \frac{2}{2} = 1.0$$

- **Case 2:** $P(A = \textbf{Yes}|B = \textbf{Yes}, E = \textbf{No})$**:**
  - Number of cases where $B =$ Yes and $E =$ No: 2.
  - Out of these, the alarm went off in 2 cases.

  Thus, we calculate:
  $$P(A = \text{Yes}|B = \text{Yes}, E = \text{No}) = \frac{2}{2} = 1.0$$

- **Case 3:** $P(A = \textbf{Yes}|B = \textbf{No}, E = \textbf{Yes})$**:**
  - Number of cases where $B =$ No and $E =$ Yes: 3.
  - Out of these, the alarm went off in 2 cases.

  Thus, we calculate:
  $$P(A = \text{Yes}|B = \text{No}, E = \text{Yes}) = \frac{2}{3} = 0.67$$

- **Case 4:** $P(A = \textbf{Yes}|B = \textbf{No}, E = \textbf{No})$**:**
  - Number of cases where $B =$ No and $E =$ No: 3.
  - Out of these, the alarm went off in 1 case.

Thus, we calculate:
$$P(A = \text{Yes}|B = \text{No}, E = \text{No}) = \frac{1}{3} = 0.33$$

Table 2: Conditional Probability Table for Alarm (A) given Burglary (B) and Earthquake (E)

Burglary (B)	**Earthquake (E)**	**Alarm (A = Yes)**	**Alarm (A = No)**
Yes	Yes	1.0	0.0
Yes	No	1.0	0.0
No	Yes	0.67	0.33
No	No	0.33	0.67

6.2.2 CPT for John Calls given Alarm $P(J|A)$

- **Case 1:** $P(J = \textbf{Yes}|A = \textbf{Yes})$
  - Number of cases where the alarm went off ($A = \text{Yes}$): 7.
  - Out of these, John called in 6 cases.

Thus, we calculate:

$$P(J = \text{Yes}|A = \text{Yes}) = \frac{6}{7} = 0.86$$

- **Case 2:** $P(J = \textbf{Yes}|A = \textbf{No})$
  - Number of cases where the alarm did not go off ($A = \text{No}$): 3.
  - Out of these, John called in 0 cases.

Thus, we calculate:

$$P(J = \text{Yes}|A = \text{No}) = \frac{0}{3} = 0.0$$

Table 3: CPT for John Calls (J) given Alarm (A)

Alarm (A)	**John Calls (J = Yes)**	**John Calls (J = No)**
Yes	0.86	0.14
No	0.0	1.0

6.2.3 CPT for Mary Calls given Alarm $P(M|A)$

- **Case 1:** $P(M = \textbf{Yes}|A = \textbf{Yes})$
  - Number of cases where the alarm went off ($A = \text{Yes}$): 7.
  - Out of these, Mary called in 5 cases.

Thus, we calculate:

$$P(M = \text{Yes}|A = \text{Yes}) = \frac{5}{7} = 0.71$$

- **Case 2:** $P(M = \textbf{Yes}|A = \textbf{No})$
  - Number of cases where the alarm did not go off ($A = \text{No}$): 3.
  - Out of these, Mary called in 0 cases.

Thus, we calculate:

$$P(M = \text{Yes}|A = \text{No}) = \frac{0}{3} = 0.0$$

Table 4: CPT for Mary Calls (M) given Alarm (A)

Alarm (A)	**Mary Calls (M = Yes)**	**Mary Calls (M = No)**
Yes	0.71	0.29
No	0.0	1.0

7 Solution

We need to calculate the joint probability:

$$P(B = \text{Yes}, E = \text{No}, A = \text{Yes}, J = \text{Yes}, M = \text{Yes})$$

Using the chain rule for conditional probabilities:

$$\begin{aligned} P(B = \text{Yes}, E = \text{No}, A = \text{Yes}, J = \text{Yes}, M = \text{Yes}) = P(B = \text{Yes}) \cdot P(E = \text{No}) \cdot P(A = \text{Yes}|B = \text{Yes}, E = \text{No}) \\ \cdot P(J = \text{Yes}|A = \text{Yes}) \cdot P(M = \text{Yes}|A = \text{Yes}) \end{aligned}$$

Substituting the values, we get:

$$\begin{aligned} P(B = \text{Yes}, E = \text{No}, A = \text{Yes}, J = \text{Yes}, M = \text{Yes}) &= 0.4 \times 0.5 \times 1.0 \times 0.86 \times 0.71 \\ &= 0.1221 \end{aligned}$$

The probability of the event that a burglary has occurred, no earthquake has occurred, the alarm has sounded, and both John and Mary called is approximately 0.1221. This means there is a **12.21% chance** of this specific combination of events happening, based on the given probabilities. ■

Bayesian Network (Practice Example)

Your objective is to predict whether a student will pass an exam based on the following features:

- **Study Hours**: How many hours the student studies per day (Low, Medium, High).
- **Attendance**: Whether the student attends classes regularly (Regular, Irregular).
- **Sleep Hours**: How many hours the student sleeps per day (Low, Medium, High).
- **Pass Exam**: Whether the student passes the exam (Yes, No).

1 Historical Database

The following table shows the raw data for 12 students:

Table 1: Training Data for Exam Passing Prediction

Student ID	Study Hours	Attendance	Sleep Hours	Pass Exam (Class)
1	Low	Irregular	Low	No
2	Medium	Regular	Medium	Yes
3	High	Regular	High	Yes
4	Medium	Regular	High	Yes
5	Low	Irregular	Low	No
6	Low	Irregular	Medium	No
7	High	Regular	High	Yes
8	Medium	Irregular	Medium	No
9	Medium	Regular	Medium	Yes
10	High	Regular	High	Yes
11	Low	Regular	Medium	No
12	Medium	Irregular	Medium	No

2 Bayesian Network Structure

The structure of the Bayesian Network is as follows:

- **Study Hours** and **Attendance** influence **Sleep Hours**.
- **Study Hours**, **Attendance**, and **Sleep Hours** together influence the probability of **Pass Exam**.

3 Bayesian Network Diagram

The diagram below represents the Bayesian Network for predicting whether a student will pass the exam based on their study habits, attendance, and sleep hours.

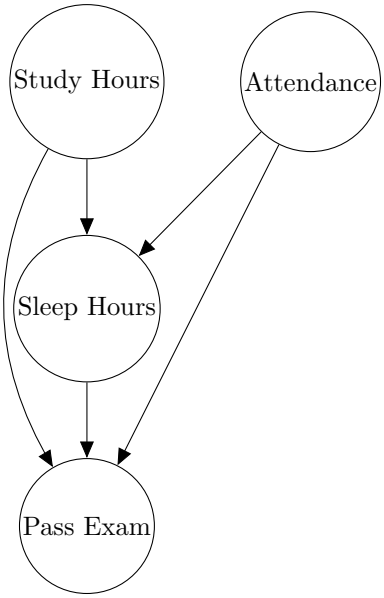


Figure 1: Bayesian Network for Exam Passing Prediction

4 Questions

1. **Construct the Bayesian Network**: Define the Conditional Probability Tables (CPTs) based on the raw database.
2. **Inference**: Calculate the probability that a student will pass the exam given:
   - Study Hours: Medium
   - Attendance: Regular
   - Sleep Hours: Medium
3. **Explain the Impact**: Explain how changing the values of Study Hours, Attendance, and Sleep Hours affects the probability of passing the exam.

■

Naïve Bayes Classification with Laplacian Correction

Let's consider a small example of a **Naïve Bayesian Classifier** where we predict whether a person will **buy a computer** based on their **income** and **student status**. The dataset is small, and in this case, we need to use **Laplacian correction** (also known as Laplace smoothing) to handle cases where we have zero counts in the conditional probabilities.

1 Dataset

We consider the following dataset where we predict whether a person will buy a computer based on their income and student status:

Table 1: Dataset for Buy Computer Class Prediction

ID	Income	Student	Buys Computer (Class)
1	High	No	No
2	High	No	No
3	Medium	No	Yes
4	Low	Yes	Yes
5	Low	Yes	Yes
6	Low	No	No

We aim to predict whether a person with *High Income* and *Student Status = Yes* will buy a computer.

2 Steps to Perform Naïve Bayes Classification

2.1 Prior Probabilities

The prior probabilities for each class are calculated as follows:

$$P(\text{Yes}) = \frac{\text{Number of people who bought a computer}}{\text{Total number of people}} = \frac{3}{6} = 0.5$$

$$P(\text{No}) = \frac{\text{Number of people who did not buy a computer}}{\text{Total number of people}} = \frac{3}{6} = 0.5$$

2.2 Likelihoods Without Laplacian Correction

Next, we calculate the likelihoods for each feature given the class.

2.2.1 Income

For the class Yes (Buys Computer):

$$P(\text{Income} = \text{High}|\text{Yes}) = \frac{0}{3} = 0$$

For the class No (Does Not Buys Computer):

$$P(\text{Income} = \text{High}|\text{No}) = \frac{2}{3} = 0.67$$

2.2.2 Student Status

For the class Yes:

$$P(\text{Student} = \text{Yes}|\text{Yes}) = \frac{2}{3} = 0.67$$

For the class No:

$$P(\text{Student} = \text{Yes}|\text{No}) = \frac{0}{3} = 0$$

2.3 Laplacian Correction

To avoid zero probabilities, we apply Laplacian correction by adding 1 to the counts.

2.3.1 Corrected Likelihoods for Income (with $k = 3$)

For the class Yes:

$$P(\text{Income} = \text{High}|\text{Yes}) = \frac{0+1}{3+3} = \frac{1}{6} = 0.167$$

For the class No:

$$P(\text{Income} = \text{High}|\text{No}) = \frac{2+1}{3+3} = \frac{3}{6} = 0.5$$

2.3.2 Corrected Likelihoods for Student Status (with $k = 2$)

For the class Yes:

$$P(\text{Student} = \text{Yes}|\text{Yes}) = \frac{2+1}{3+2} = \frac{3}{5} = 0.6$$

For the class No:

$$P(\text{Student} = \text{Yes}|\text{No}) = \frac{0+1}{3+2} = \frac{1}{5} = 0.2$$

2.4 Posterior Probabilities

Using the corrected likelihoods, we calculate the posterior probabilities for each class using the Naïve Bayes formula.

2.4.1 For Class = Yes

$$P(\text{Yes}|\text{High Income, Student} = \text{Yes}) = P(\text{Yes}) \cdot P(\text{Income} = \text{High}|\text{Yes}) \cdot P(\text{Student} = \text{Yes}|\text{Yes})$$
$$= 0.5 \times 0.167 \times 0.6 = 0.0501$$

2.4.2 For Class = No

$$P(\text{No}|\text{High Income, Student} = \text{Yes}) = P(\text{No}) \cdot P(\text{Income} = \text{High}|\text{No}) \cdot P(\text{Student} = \text{Yes}|\text{No})$$
$$= 0.5 \times 0.5 \times 0.2 = 0.05$$

2.5 Final Prediction

Since $P(\text{Yes}|\text{High Income, Student} = \text{Yes}) = 0.0501$ is greater than $P(\text{No}|\text{High Income, Student} = \text{Yes}) = 0.05$, the model predicts that the person will **buy a computer**. ■

Effect of Using k in Laplacian Correction for Naïve Bayesian Classification

In Naïve Bayesian classification, Laplacian correction (also known as Laplace smoothing) is used to handle zero probabilities when a feature value does not appear in the training data for a particular class. The correction prevents the entire probability from becoming zero when multiplying the probabilities of feature values. A key aspect of Laplacian correction is the use of k, the number of possible values a feature can take. This document demonstrates the effect of not using k in Laplacian correction and how it can result in incorrect probabilities.

1 Problem Statement

Consider a feature called *Weather* with three possible values:

- *Sunny*
- *Rainy*
- *Cloudy*

We are predicting whether people will go *Outdoors* (Yes/No) based on the weather. The training dataset is shown below:

Table 1: Training Dataset

ID	Weather	Outdoors (Class)
1	Sunny	Yes
2	Sunny	Yes
3	Rainy	Yes
4	Sunny	No
5	Rainy	No
6	Sunny	No

1.1 Without Laplacian Correction

The probabilities for the weather conditions given "Outdoors = Yes" before any smoothing are:

$$P(\textit{Sunny} \mid \textit{Yes}) = \frac{2}{3}, \quad P(\textit{Rainy} \mid \textit{Yes}) = \frac{1}{3}, \quad P(\textit{Cloudy} \mid \textit{Yes}) = 0$$

The sum of the probabilities is:

$$P(\textit{Sunny} \mid \textit{Yes}) + P(\textit{Rainy} \mid \textit{Yes}) + P(\textit{Cloudy} \mid \textit{Yes}) = \frac{2}{3} + \frac{1}{3} + 0 = 1$$

However, the zero probability for *Cloudy* will cause problems if this value is encountered during classification.

1.2 With Laplacian Correction (Without k)

Using Laplacian correction without k (i.e., adding 1 to the numerator and denominator), we calculate the probabilities as:

$$P(\textit{Sunny} \mid \textit{Yes}) = \frac{2+1}{3+1} = \frac{3}{4}, \quad P(\textit{Rainy} \mid \textit{Yes}) = \frac{1+1}{3+1} = \frac{2}{4} = 0.5$$

$$P(\textit{Cloudy} \mid \textit{Yes}) = \frac{0+1}{3+1} = \frac{1}{4}$$

The sum of the probabilities becomes:

$$P(\textit{Sunny} \mid \textit{Yes}) + P(\textit{Rainy} \mid \textit{Yes}) + P(\textit{Cloudy} \mid \textit{Yes}) = \frac{3}{4} + 0.5 + \frac{1}{4} = \frac{6}{4} = 1.5$$

This is greater than 1, which violates the rules of probability.

1.3 With Laplacian Correction (With $k = 3$)

Now, using Laplacian correction with $k = 3$ (since there are three possible values for *Weather*: Sunny, Rainy, Cloudy), we recalculate the probabilities:

$$P(Sunny \mid Yes) = \frac{2+1}{3+3} = \frac{3}{6} = 0.5$$

$$P(Rainy \mid Yes) = \frac{1+1}{3+3} = \frac{2}{6} = \frac{1}{3}$$

$$P(Cloudy \mid Yes) = \frac{0+1}{3+3} = \frac{1}{6}$$

The sum of the probabilities becomes:

$$P(Sunny \mid Yes) + P(Rainy \mid Yes) + P(Cloudy \mid Yes) = 0.5 + \frac{1}{3} + \frac{1}{6} = \frac{6}{6} = 1$$

2 Conclusion

The use of k in Laplacian correction is essential for maintaining the sum of probabilities as 1. Without k, the smoothing effect is not properly distributed across all possible values of the feature, leading to overestimated probabilities. By including k, we ensure that the probabilities are correctly normalized and that unseen values do not disproportionately influence the classification. ■

Naïve Bayes Classification with Laplacian Correction (Practice Example)

1 Problem Description

You are tasked with building a Naive Bayesian classifier to predict whether a student will pass or fail a course based on the following features:

- **Attendance**: (Low, Medium, High)
- **Study Hours**: (Low, Medium, High)
- **Previous Grades**: (Pass, Fail)

The dataset is as follows:

Table 1: Dataset for Pass/Fail Prediction

ID	Attendance	Study Hours	Previous Grades	Pass/Fail (Class)
1	Low	Low	Fail	Fail
2	High	High	Pass	Pass
3	Medium	Low	Pass	Fail
4	Medium	Medium	Fail	Fail
5	High	Medium	Pass	Pass
6	Low	High	Fail	Fail
7	Medium	Medium	Pass	Pass
8	Low	Low	Fail	Fail
9	High	High	Pass	Pass
10	Medium	Low	Fail	Fail

2 Questions

1. **Calculate the Prior Probabilities** for both classes (Pass/Fail).
2. **Calculate the Conditional Probabilities** for each feature given the class, using **Laplacian Correction** to handle any zero probabilities.
3. **Classify the following student using the Naive Bayesian algorithm:**
   - **Attendance**: Medium
   - **Study Hours**: Medium
   - **Previous Grades**: Pass
4. **Compare the result** with the original dataset and discuss how Laplacian correction helps prevent issues with zero probabilities.

3 Expected Learning Outcomes

Students will learn how to handle small datasets with zero probabilities by applying Laplacian correction.

Naïve Bayes Classification for Continuous Values with Laplace Correction

In a **Naïve Bayesian Classification** scenario with continuous values like income, you need to treat the continuous attributes differently from discrete ones. The **Gaussian Naïve Bayes** model is commonly used for handling continuous features, where each continuous feature is assumed to follow a **normal (Gaussian) distribution**.

1 Problem Statement

We will predict whether a person will **buy a computer** based on their:

- **Age** (categorical)
- **Income** (continuous)
- **Student status** (categorical)

2 Dataset

Table 1: Dataset for Buy Computer Class Prediction

ID	Age	Income	Student	Buys Computer (Class)
1	Youth	300	No	No
2	Youth	400	No	No
3	Middle	100	No	Yes
4	Senior	300	Yes	Yes
5	Senior	200	Yes	Yes
6	Youth	300	Yes	No

We want to predict whether a new person with:

- **Age = Senior**
- **Income = 400**
- **Student Status = Yes**

3 Steps to Perform Naïve Bayes Classification

3.1 Calculate Prior Probabilities

We first calculate the prior probabilities for the target class, i.e., the probability of buying a computer or not.

$$P(\text{Yes}) = \frac{\text{Number of people who bought a computer}}{\text{Total number of people}} = \frac{3}{6} = 0.5$$

$$P(\text{No}) = \frac{3}{6} = 0.5$$

3.2 Likelihood for Categorical Attributes (Age and Student Status)

We calculate the likelihoods for the **categorical** attributes (Age and Student status) based on the class.

3.2.1 Age:

Since we have three categories for the Age attribute (**Youth**, **Middle**, and **Senior**), we set $k = 3$.

-For **Class = Yes:**

There are 2 people with **Age = Senior** who bought a computer. Applying Laplacian correction:

$$P(\text{Age} = \text{Senior}|\text{Yes}) = \frac{\text{Count of Senior and Yes} + 1}{\text{Total Yes instances} + k}$$

$$P(\text{Age} = \text{Senior}|\text{Yes}) = \frac{2+1}{3+3} = \frac{3}{6} = 0.5$$

-For **Class = No:**

There are zero people with **Age = Senior** who did not buy a computer. Applying Laplacian correction:

$$P(\text{Age} = \text{Senior}|\text{No}) = \frac{\text{Count of Senior and No} + 1}{\text{Total No instances} + k}$$

$$P(\text{Age} = \text{Senior}|\text{No}) = \frac{0+1}{3+3} = \frac{1}{6} = 0.167$$

3.2.2 Student Status:

- For **Student = Yes** given **Yes**:

$$P(\text{Student} = \text{Yes}|\text{Yes}) = \frac{2}{3} = 0.67$$

- For **Student = Yes** given **No**:

$$P(\text{Student} = \text{Yes}|\text{No}) = \frac{1}{3} = 0.33$$

3.3 Likelihood for Continuous Attribute (Income)

For continuous attributes like **Income**, we assume that they follow a **Gaussian (normal) distribution**. The probability density function for a Gaussian distribution is:

$$P(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Where:

- x is the income value for the new instance (in this case, 400).
- μ is the mean income for the class.
- σ is the standard deviation of income for the class.

Let's calculate the likelihood of **Income = 400** given the class.

For Class = Yes:

- Mean (μ_{Yes}) of income for class **Yes**:

$$\mu_{\text{Yes}} = \frac{300 + 200 + 100}{3} = 200$$

- Standard deviation (σ_{Yes}) of income for class **Yes**:

$$\sigma_{\text{Yes}} = \sqrt{\frac{(300-200)^2 + (200-200)^2 + (100-200)^2}{3}} = \sqrt{\frac{10000 + 0 + 10000}{3}} = 100$$

Now, we calculate the likelihood of **Income = 400** given **Yes**:

$$P(\text{Income} = 400|\text{Yes}) = \frac{1}{\sqrt{2\pi(100)^2}} \exp\left(-\frac{(400-200)^2}{2(100)^2}\right)$$

$$= \frac{1}{\sqrt{2\pi(100)^2}} \exp\left(-\frac{(200)^2}{20000}\right) = \frac{1}{\sqrt{62831.85}} \exp(-2)$$

$$P(\text{Income} = 400|\text{Yes}) \approx 0.0054$$

3.3.1 For Class = No:

- Mean (μ_{No}) of income for class **No**:

$$\mu_{\text{No}} = \frac{300 + 400 + 300}{3} = 333.33$$

- Standard deviation (σ_{No}) of income for class **No**:

$$\sigma_{\text{No}} = \sqrt{\frac{(300 - 333.33)^2 + (400 - 333.33)^2 + (300 - 333.33)^2}{3}} = \sqrt{\frac{1111.11 + 4444.44 + 1111.11}{3}} = 47.14$$

Now, we calculate the likelihood of **Income = 400** given **No**:

$$P(\text{Income} = 400|\text{No}) = \frac{1}{\sqrt{2\pi(47.14)^2}} \exp\left(-\frac{(400 - 333.33)^2}{2(47.14)^2}\right)$$

$$= \frac{1}{\sqrt{2\pi(47.14)^2}} \exp\left(-\frac{(66.67)^2}{2(47.14)^2}\right) = \frac{1}{\sqrt{13948.41}} \exp(-1)$$

$$P(\text{Income} = 400|\text{No}) \approx 0.0053$$

3.4 Calculate Posterior Probabilities

Now we calculate the **posterior probabilities** for both classes using the Naïve Bayes formula:

- **For Class = Yes**:

$$\begin{aligned} P(\text{Yes}|\text{Age=Senior}, \text{Income} = 400, \text{Student} = \text{Yes}) &= P(\text{Yes}) \cdot P(\text{Age} = \text{Senior}|\text{Yes}) \\ &\quad \cdot P(\text{Income} = 400|\text{Yes}) \cdot P(\text{Student} = \text{Yes}|\text{Yes}) \\ &= 0.5 \times 0.5 \times 0.0054 \times 0.67 \\ &= 0.00090 \end{aligned}$$

- **For Class = No**:

$$\begin{aligned} P(\text{No}|\text{Age=Senior}, \text{Income} = 400, \text{Student} = \text{Yes}) &= P(\text{No}) \cdot P(\text{Age} = \text{Senior}|\text{No}) \\ &\quad \cdot P(\text{Income} = 400|\text{No}) \cdot P(\text{Student} = \text{Yes}|\text{No}) \\ &= 0.5 \times 0.167 \times 0.0053 \times 0.33 \\ &= 0.00015 \end{aligned}$$

3.5 Final Prediction

Comparing the posterior probabilities:
- $P(\text{Yes}) = 0.00090$
- $P(\text{No}) = 0.00015$

Since $P(\text{Yes}) > P(\text{No})$, the model predicts that the person **will buy a computer**.

4 Conclusion

For continuous attributes like **Income**, the Gaussian Naïve Bayes classifier is used, and for categorical attributes, standard probability calculations are performed. **Laplacian correction** is applied where needed to avoid zero probabilities. ■

Naïve Bayes Classification with Continuous Values (Practice Example)

1 Problem Description

You are given a dataset that contains the following attributes for employees of a company:

- **Age** (continuous)
- **Salary** (continuous)
- **Education Level** (High School, Bachelor's, Master's)
- **Buys Car** (Yes/No)

The dataset is as follows:

Table 1: Dataset for Car Purchase Prediction

ID	Age	Salary	Education Level	Buys Car (Class)
1	25	30000	Bachelor's	No
2	35	50000	Master's	Yes
3	45	70000	Bachelor's	Yes
4	28	32000	High School	No
5	55	60000	Master's	Yes
6	33	40000	High School	No
7	38	55000	Bachelor's	Yes
8	30	35000	Bachelor's	No
9	50	65000	Master's	Yes
10	40	52000	Bachelor's	Yes

2 Questions

1. **Calculate the Prior Probabilities** for the class "Buys Car" (Yes/No).
2. For the continuous attributes **Age** and **Salary**, assume they follow a **Gaussian distribution**. For each class:
   - Calculate the **mean** and **standard deviation** for **Age** and **Salary**.
3. **Classify the following employee using the Gaussian Naive Bayesian algorithm:**
   - **Age**: 37
   - **Salary**: 48000
   - **Education Level**: Bachelor's
4. Show your calculations for the **probability density function** of the Gaussian distribution for continuous values of **Age** and **Salary**.
5. Compare the final result with the original dataset and discuss how continuous features are handled differently from categorical features in Naive Bayes.

3 Expected Learning Outcomes

Students will gain experience in handling continuous attributes using the Gaussian Naïve Bayes approach and understand how to classify data with continuous features.

Hidden Markov Model Evaluation Problem using Forward Algorithm

In this document, we will solve the **evaluation problem** using the **forward algorithm** for a given Hidden Markov Model (HMM). The problem we are solving is to find the probability of the observation sequence: {Walk, Shop, Clean}

1 Historical Data

The table below shows the historical data of 10 days, with hidden weather states (Sunny or Rainy) and observable activities (Walk, Shop, Clean).

Table 1: Historical Data of Weather (Hidden States) and Activities (Observations)

Day	Weather (Hidden State)	Activity (Observation)
1	Sunny	Walk
2	Sunny	Shop
3	Rainy	Clean
4	Sunny	Walk
5	Rainy	Shop
6	Rainy	Clean
7	Sunny	Clean
8	Rainy	Walk
9	Sunny	Shop
10	Rainy	Clean

We are given a Hidden Markov Model (HMM) for weather prediction with the following parameters:

- **Hidden States:** $S = \{\text{Rainy}, \text{Sunny}\}$
- **Observations:** $O = \{\text{Walk}, \text{Shop}, \text{Clean}\}$
- **Transition Probabilities:**

  The transition matrix A for a Hidden Markov Model is defined as:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1N} \\ a_{21} & a_{22} & \cdots & a_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ a_{N1} & a_{N2} & \cdots & a_{NN} \end{bmatrix}$$

  Where:

  - $a_{ij} = P(S_t = j \mid S_{t-1} = i)$, represents the probability of transitioning from state i to state j,
  - S_t is the hidden state at time step t,
  - N is the number of hidden states.

  The sum of each row in the transition matrix must equal 1:

$$\sum_{j=1}^{N} a_{ij} = 1, \quad \forall i$$

  Thus, the transition matrix A shows the probability of moving from one hidden weather state (Sunny or Rainy) to another.

$$A = \begin{bmatrix} P(\text{Sunny}|\text{Sunny}) = \frac{1}{5} & P(\text{Rainy}|\text{Sunny}) = \frac{4}{5} \\ P(\text{Sunny}|\text{Rainy}) = \frac{3}{4} & P(\text{Rainy}|\text{Rainy}) = \frac{1}{4} \end{bmatrix}$$

  This results in the following transition matrix:

$$A = \begin{bmatrix} 0.20 & 0.80 \\ 0.75 & 0.25 \end{bmatrix}$$

- **Emission Probabilities:**

  The emission matrix B for a Hidden Markov Model is defined as:

$$B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1M} \\ b_{21} & b_{22} & \cdots & b_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ b_{N1} & b_{N2} & \cdots & b_{NM} \end{bmatrix}$$

  Where:

  - $b_{ij} = P(o_j \mid S_i)$, represents the probability of observing o_j (the j-th observation) given the hidden state S_i (the i-th hidden state),
  - S_i is the hidden state at time t,
  - o_j is an observation,
  - N is the number of hidden states,
  - M is the number of possible observations.

  The sum of each row in the emission matrix must equal 1:

$$\sum_{j=1}^{M} b_{ij} = 1, \quad \forall i$$

  Thus, the emission matrix B represents the probabilities of observing activities (Walk, Shop, Clean) given a weather state (Sunny or Rainy).

$$B = \begin{bmatrix} P(\text{Walk}|\text{Sunny}) = \frac{2}{5} & P(\text{Shop}|\text{Sunny}) = \frac{2}{5} & P(\text{Clean}|\text{Sunny}) = \frac{1}{5} \\ P(\text{Walk}|\text{Rainy}) = \frac{1}{5} & P(\text{Shop}|\text{Rainy}) = \frac{1}{5} & P(\text{Clean}|\text{Rainy}) = \frac{3}{5} \end{bmatrix}$$

  This results in the following emission matrix:

$$B = \begin{bmatrix} 0.4 & 0.4 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

- **Initial Probabilities:**

$$\pi = \{P(\text{Sunny}) = 0.5, P(\text{Rainy}) = 0.5\}$$

We want to compute the probability of the observation sequence $O = \{\text{Walk, Shop, Clean}\}$ using the forward algorithm.

2 Solution

2.1 Initialization

The initialization step involves calculating the forward probabilities for each hidden state at time $t = 1$, which corresponds to the first observation (*Walk*). The formula for initialization is:

$$\alpha_1(i) = \pi_i \cdot b_i(o_1)$$

Where:

- $\alpha_1(i)$ is the forward probability for state i at time $t = 1$,
- π_i is the initial probability of state i,
- $b_i(o_1)$ is the emission probability of observing o_1 in state i.

Using the initial probabilities and emission probabilities for the first observation (*Walk*), we calculate:

$$\alpha_1(\text{Sunny}) = \pi(\text{Sunny}) \cdot P(\text{Walk}|\text{Sunny}) = 0.5 \times 0.4 = 0.2$$

$$\alpha_1(\text{Rainy}) = \pi(\text{Rainy}) \cdot P(\text{Walk}|\text{Rainy}) = 0.5 \times 0.2 = 0.1$$

2.2 Recursion

For each subsequent time step t, the forward probabilities are recursively calculated using the formula:

$$\alpha_t(j) = \sum_{i=1}^{N} \alpha_{t-1}(i) \cdot a_{ij} \cdot b_j(o_t)$$

Where:

- $\alpha_t(j)$ is the forward probability for state j at time t,
- $\alpha_{t-1}(i)$ is the forward probability for state i at time $t-1$,
- a_{ij} is the transition probability from state i to state j,
- $b_j(o_t)$ is the emission probability of observing o_t in state j.

At $t = 2$ (second observation: *Shop*), we calculate:

$$\alpha_2(\text{Sunny}) = [\alpha_1(\text{Sunny}) \cdot P(\text{Sunny} \mid \text{Sunny}) + \alpha_1(\text{Rainy}) \cdot P(\quad)] \cdot P(\text{Shop} \mid \text{Sunny})$$
$$\begin{aligned}\alpha_2(\text{Sunny}) &= [0.2 \cdot 0.20 + 0.1 \cdot 0.80] \cdot 0.4 \\ &= [0.04 + 0.08] \cdot 0.4 \\ &= 0.12 \cdot 0.4 = 0.048\end{aligned}$$

$$\alpha_2(\text{Rainy}) = [\alpha_1(\text{Sunny}) \cdot P(\quad) + \alpha_1(\text{Rainy}) \cdot P(\text{Rainy} \mid \text{Rainy})] \cdot P(\text{Shop} \mid \text{Rainy})$$
$$\begin{aligned}\alpha_2(\text{Rainy}) &= [0.2 \cdot 0.75 + 0.1 \cdot 0.25] \cdot 0.2 \\ &= [0.15 + 0.025] \cdot 0.2 \\ &= 0.175 \cdot 0.2 = 0.035\end{aligned}$$

At $t = 3$ (third observation: *Clean*), we calculate:

$$\alpha_3(\text{Sunny}) = [\alpha_2(\text{Sunny}) \cdot P(\text{Sunny} \mid \text{Sunny}) + \alpha_2(\text{Rainy}) \cdot P(\text{Sunny} \mid \text{Rainy})] \cdot P(\text{Clean} \mid \text{Sunny})$$
$$\begin{aligned}\alpha_3(\text{Sunny}) &= [0.048 \cdot 0.20 + 0.035 \cdot 0.80] \cdot 0.2 \\ &= [0.0096 + 0.028] \cdot 0.2 \\ &= 0.0376 \cdot 0.2 = 0.00752\end{aligned}$$

$$\alpha_3(\text{Rainy}) = [\alpha_2(\text{Sunny}) \cdot P(\text{Rainy} \mid \text{Sunny}) + \alpha_2(\text{Rainy}) \cdot P(\text{Rainy} \mid \text{Rainy})] \cdot P(\text{Clean} \mid \text{Rainy})$$
$$\begin{aligned}\alpha_3(\text{Rainy}) &= [0.048 \cdot 0.75 + 0.035 \cdot 0.25] \cdot 0.6 \\ &= [0.036 + 0.00875] \cdot 0.6 \\ &= 0.04475 \cdot 0.6 = 0.02685\end{aligned}$$

2.3 Termination

The total probability of observing the sequence is calculated by summing the forward probabilities at the final time step T:

$$P(O \mid \lambda) = \sum_{i=1}^{N} \alpha_T(i)$$

Where:

- $P(O \mid \lambda)$ is the total probability of the observation sequence O,
- $\alpha_T(i)$ is the forward probability for state i at the final time step T.

Thus, the final step involves summing the forward probabilities at time $t = 3$ to obtain the total probability of the observation sequence $O = \{\text{Walk, Shop, Clean}\}$:

$$P(O|\lambda) = \alpha_3(\text{Sunny}) + \alpha_3(\text{Rainy}) = 0.00752 + 0.02685 = 0.03437$$

Thus, the probability of observing the sequence **Walk, Shop, Clean** given the HMM is approximately $P(O|\lambda) = 0.03437$.

3 Conclusion

In this example, we used the forward algorithm to calculate the probability of the observation sequence *Walk, Shop, Clean* using the Hidden Markov Model's transition and emission matrices. The result shows how likely this sequence is under the given HMM parameters. ■

Practice Example: Hidden Markov Model (HMM) Sequence Prediction (Human Mood and Activities)

1 Problem Description

Your objective is to predict the probability of an observation sequence based on a person's mood (hidden states) and the activities they perform. The possible moods are **Happy** and **Sad**, and the activities people perform are **Exercise**, **Read**, and **Watch TV**.

Your goal is to:

1. Calculate the **transition matrix** for the hidden states (Happy and Sad).
2. Calculate the **emission matrix** for the observed activities (Exercise, Read, Watch TV).
3. Use the **forward algorithm** to calculate the probability of a given observation sequence.

2 Historical Data

The following table represents 12 days of human mood conditions (hidden states) and observed activities.

Table 1: Raw Data of Moods and Activities

Day	Mood (Hidden State)	Activity (Observation)
1	Happy	Exercise
2	Sad	Watch TV
3	Happy	Read
4	Happy	Exercise
5	Sad	Watch TV
6	Sad	Read
7	Happy	Exercise
8	Sad	Watch TV
9	Happy	Exercise
10	Happy	Read
11	Sad	Watch TV
12	Happy	Exercise

3 Questions

3.1 Calculate the Transition Matrix

The transition matrix represents the probability of moving from one mood to another between consecutive days (e.g., from Happy to Sad).

3.2 Calculate the Emission Matrix

The emission matrix represents the probability of observing each activity (Exercise, Read, Watch TV) given a particular mood (Happy or Sad).

3.3 Predict the Probability of an Observation Sequence

Given the observation sequence: $O = (\text{Exercise}, \text{Read}, \text{Watch TV})$, use the forward algorithm to calculate the probability that this sequence was generated by the HMM with the transition and emission matrices you calculated.

Instructions

- **Transition Matrix Calculation**: Count the transitions from Happy to Happy, Happy to Sad, Sad to Happy, and Sad to Sad from the raw data.
- **Emission Matrix Calculation**: Count how many times each activity (Exercise, Read, Watch TV) occurred given the mood of the person.
- **Forward Algorithm**: Use the forward algorithm step-by-step to calculate the final probability of the observation sequence $O = (\text{Exercise}, \text{Read}, \text{Watch TV})$.

Hints

- For the **transition matrix**, the probabilities should sum to 1 for each row (from a given mood).
- For the **emission matrix**, the probabilities should sum to 1 for each mood.
- Use the forward algorithm formulas to compute the final probability.

■